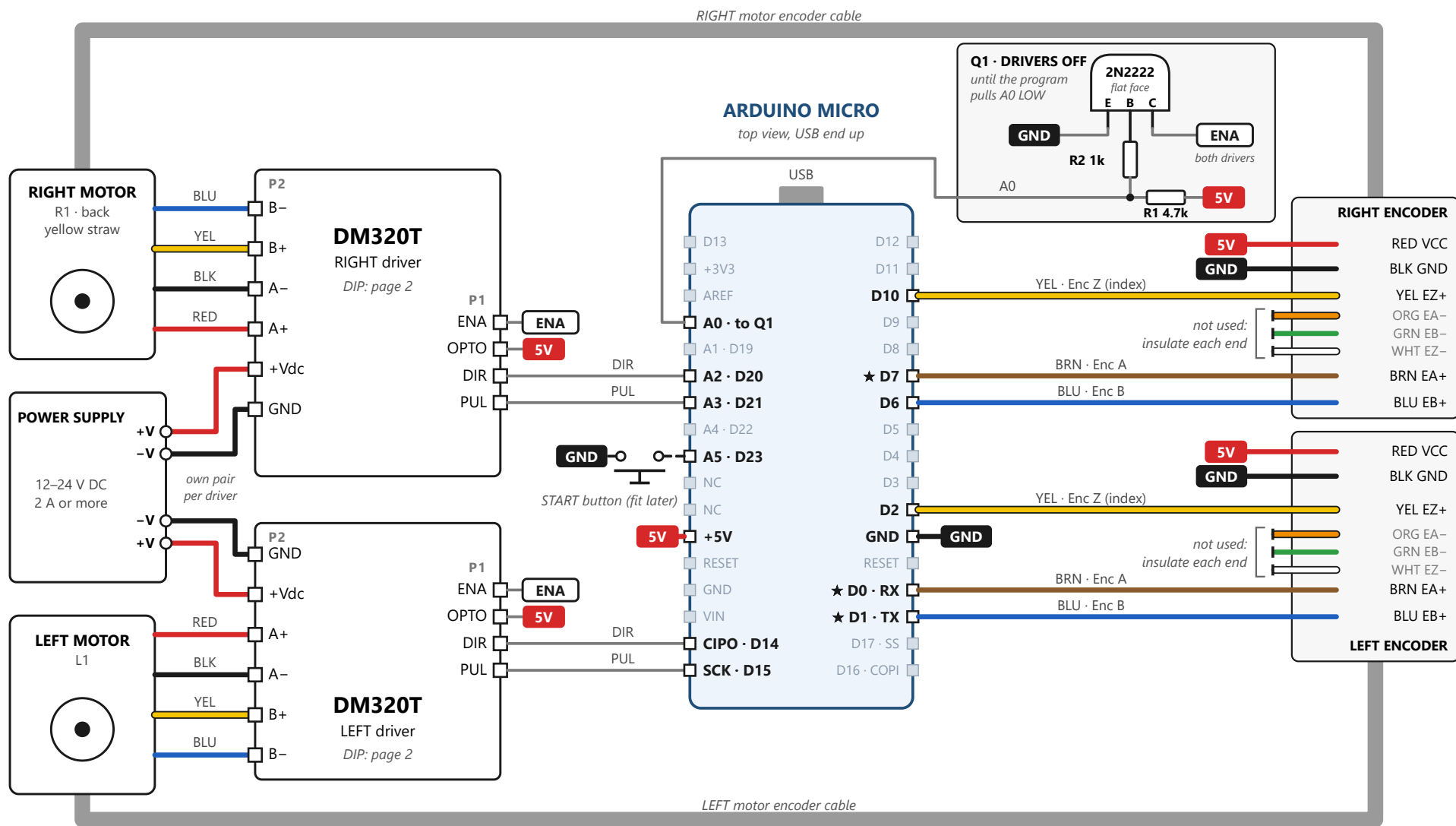


Triangle maquette: wiring

Arduino Micro · 2 × DM320T · 2 × 17HS08-1004-ME1K (with encoder) · v7, 1 Oct 2026 · **carrier-board pins: set CARRIER_BOARD true in config.h**



— **Thin grey** = hook-up wire, any colour
 — **Thick** = wire colour as labelled (RED, BLK...)
 ★ = interrupt pin (encoder A, and the left encoder's B)
Dashed = fit later: the START button (A5).
 Micro drawn from above, USB end up, pins in real header order. Greyed pins are unused.

Tags: every **5V** joins together, every **GND** joins together, and both drivers' **ENA** go to Q1's C. The Micro has one +5V pin (left-hand header), so split it 5 ways (2 × OPTO, 2 × encoder RED, R1) with a terminal strip or lever connectors. The encoder BLK wires and Q1's E go to the GND pin on the right-hand header (either GND works). The PSU –V does **not** connect to the Micro.

⚠ Never plug or unplug the green terminal blocks while powered.

Keep PUL/DIR and encoder wires ≥ 10 cm from the motor leads. The motor and encoder cables both use RED/BLK/YEL/BLU, so label them.

Connection checklist and driver settings

Tick each connection as you make it

LEFT motor (L1)

MOTOR LEADS → LEFT DRIVER P2

☐ RED	→ A+
☐ BLK	→ A−
☐ YEL	→ B+
☐ BLU	→ B−

MICRO → LEFT DRIVER P1

☐ D15 · SCK	→ PUL
☐ D14 · CIPO	→ DIR
☐ 5V	→ OPTO
☐ Q1 C	→ ENA

ENCODER CABLE → MICRO

☐ RED · VCC	→ 5V
☐ BLK · GND	→ GND
☐ BRN · EA+	→ D0 ★
☐ BLU · EB+	→ D1 ★
☐ YEL · EZ+	→ D2
☐ ORG, GRN, WHT	→ insulate, not connected

Pins in main.cpp

```
// config.h, CARRIER_BOARD true
AccelStepper stepperLeft (DRIVER, 15, 14); // PUL, DIR
AccelStepper stepperRight(DRIVER, A3, A2); // PUL, DIR
Encoder encoderLeft (0, 1); // index Z: D2
Encoder encoderRight(7, 6); // index Z: D10
DRIVER_ENABLE = A0; // LOW = drivers on, via Q1
// START button A5 (not wired yet)
```

RIGHT motor (R1, back, yellow straw)

MOTOR LEADS → RIGHT DRIVER P2

☐ RED	→ A+
☐ BLK	→ A−
☐ YEL	→ B+
☐ BLU	→ B−

MICRO → RIGHT DRIVER P1

☐ A3 · D21	→ PUL
☐ A2 · D20	→ DIR
☐ 5V	→ OPTO
☐ Q1 C	→ ENA

ENCODER CABLE → MICRO

☐ RED · VCC	→ 5V
☐ BLK · GND	→ GND
☐ BRN · EA+	→ D7 ★
☐ BLU · EB+	→ D6
☐ YEL · EZ+	→ D10
☐ ORG, GRN, WHT	→ insulate, not connected

Notes

- **No resistors on PUL/DIR.** The DM320T inputs are opto-isolated, wired common-anode: OPTO to +5 V, and the Micro pulls PUL and DIR low (manual fig. 2).
- **Q1 holds the drivers off** (motors free, no jerk) whenever A0 isn't driven: at power-on, uploads and resets. The program pulls A0 LOW on r. Check: before r, the shafts turn by hand.
- **Supply.** A separate pair of wires from the PSU to each driver; 12 V is quieter at slow speeds. The PSU ground stays separate from the Micro (the opto inputs isolate them). Fit the mains cover.

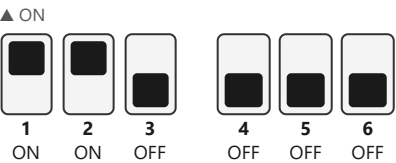
Power

☐ PSU +V	→ Left driver +Vdc
☐ PSU −V	→ Left driver GND
☐ PSU +V	→ Right driver +Vdc
☐ PSU −V	→ Right driver GND
☐ Micro USB	→ PC or USB charger
☐ A5	→ START button → GND (fit later)

Q1 · 2N2222, FLAT FACE TOWARDS YOU: E B C

☐ Q1 E	→ GND
☐ Q1 B	→ R2 1k → A0
☐ R1 4.7k	→ A0 to 5V
☐ Q1 C	→ both ENA (rows left)

DIP switches (both drivers the same)



SW1–3 Current 1.3 A peak (0.92 A RMS)
Motor is rated 1.0 A; the manual says × 1.4

SW4–6 8000 steps/rev (40 microsteps)
Exactly 2 steps per encoder count

Change switches only with power off.

- **Encoder supply** 3.3–5.5 V, so the Micro's 5 V is fine. The outputs are 5 V logic, straight to the Micro.
- **Unused encoder wires** (ORG, GRN, WHT) are driven outputs: insulate each one separately. Don't ground them.
- **Driver red LED:** 1 flash = over-current, 2 = over-voltage (> 34 V). Power off and on to reset.
- **△ Same pins as the carrier board (board.pdf), so this tests the board's pin map before it's built. Recheck directions with the test sketch after rewiring.**